ANSWER KEY & MARKING SCHEME — SCIENCE · CLASS 10 · SET 02

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — BIOLOGY — 16 questions · 30 marks

Q1–Q7 MCQs. Q8–Q9: Assertion (A) / Reason (R) – choose A, B, C or D.

1. Which of the following shows the correct path of air in the human respiratory system?

A. nostrils → pharynx → larynx → trachea → bronchi → bronchioles → alveoli
B. nostrils → larynx → pharynx → trachea → bronchioles → bronchi → alveoli
C. nostrils → pharynx → trachea → larynx → bronchi → alveoli → bronchioles
D. mouth → trachea → pharynx → larynx → bronchi → alveoli → bronchioles

> **Ans: A** Air enters through the nostrils, passes through the pharynx, larynx, trachea, bronchi and bronchioles and reaches the alveoli.

2. After a road accident, Meena can think and speak normally, but she cannot walk in a straight line and cannot pick up a pen accurately. Which part of her brain is most likely damaged?

A. cerebrum B. cerebellum C. medulla D. hypothalamus

> **Ans: B** The cerebellum controls posture, balance and precision of voluntary movements.

3. Which of the following organism–mode of reproduction pairs is correctly matched?

A. Yeast – budding
B. Rhizopus – fragmentation
C. Planaria – spore formation
D. Plasmodium – regeneration

> **Ans: A** Yeast reproduces by budding. Rhizopus – spores; Planaria – regeneration; Plasmodium – multiple fission.

4. A heterozygous tall pea plant (Tt) is crossed with a dwarf pea plant (tt). The percentage of dwarf plants expected in the progeny is:

A. 25% B. 50% C. 75% D. 100%

> **Ans: B** Tt × tt → Tt, Tt, tt, tt: half the offspring are dwarf.

5. DDT was sprayed on a pond to kill mosquito larvae. After some years its highest concentration is likely to be found in the body of:

A. phytoplankton
B. zooplankton
C. small fish
D. fish-eating birds

> **Ans: D** Non-biodegradable chemicals get concentrated at each trophic level (biological magnification) and are highest in the top consumer.

6. Which of the following statements about the human heart is correct?

A. The right ventricle pumps oxygenated blood to the body
B. The left atrium receives oxygenated blood from the lungs
C. The left ventricle receives blood from the vena cava
D. The right atrium receives blood from the pulmonary vein

> **Ans: B** The pulmonary veins bring oxygenated blood from the lungs to the left atrium.

7. The ozone layer is being depleted mainly by the chemicals released from refrigerators and aerosol sprays. These chemicals belong to the group of:

A. hydrocarbons
B. chlorofluorocarbons
C. nitrates
D. sulphates

> **Ans: B** CFCs release chlorine atoms in the stratosphere that break down ozone.

8. **Assertion (A):** A shoot of a plant bends towards light.
Reason (R): Auxin diffuses towards the shady side of the shoot and stimulates the cells there to grow longer.

> **Ans: A** This differential growth causes phototropic bending.

9. **Assertion (A):** Plants have no nervous system but they respond to stimuli.
Reason (R): Plants use chemical (hormonal) means and changes in turgor of cells to respond to stimuli.

> **Ans: A** R explains how plants respond without nerves.

10. State two features of the alveoli that make them suitable for exchange of gases.

> Thin walls, one cell thick, and a very large surface area (millions of alveoli) (1); rich supply of blood capillaries that carry the gases (1).

11. Differentiate between pollination and fertilisation.

> Pollination: transfer of pollen from anther to stigma. Fertilisation: fusion of the male and female gametes to form a zygote (1 + 1).

OR

What changes occur in the ovary and in the ovule of a flower after fertilisation?

> The ovary ripens into the fruit (1) and the ovule develops into the seed (1).

12. Name two harmful effects of the depletion of the ozone layer on living organisms and the chemical responsible for it.

> More UV radiation reaches the earth causing skin cancer, cataract and damage to crops/aquatic life (any two, 1); chemicals – chlorofluorocarbons (CFCs) (1).

13. How is food transported in plants? Name the process and the tissue involved, and explain why energy is required for the process.

> Translocation (1) through the phloem (1). Sucrose is loaded into the phloem using energy from ATP; this raises the osmotic pressure so water enters the tissue and moves the material to areas of low pressure (1).

14. What is a reflex arc? Write two differences between a reflex action and a voluntary action.

> Reflex arc: the pathway followed by an impulse in a reflex action – receptor → sensory neuron → spinal cord → motor neuron → effector (1). Reflex action: automatic, involuntary, not controlled by the brain, fast; voluntary action: controlled by the forebrain, deliberate, slower (any two, 1 each).

15. A pond ecosystem has the food chain: phytoplankton → zooplankton → small fish → large fish → fish-eating birds. The phytoplankton in the pond fix 20 000 J of energy. The pond water was once contaminated with a pesticide, and years later the birds were found to have a very high concentration of the pesticide in their bodies.

(a) Name the producer and one tertiary consumer in the food chain.

> Producer: phytoplankton (½); tertiary consumer: large fish (½).

(b) Why did the birds have the highest concentration of the pesticide? Name this phenomenon.

> Non-biodegradable pesticide accumulates at every trophic level; the top consumer receives the most – biological magnification (1).

(c) Using the 10% law calculate the energy available to the large fish.

> Zooplankton 2000 J → small fish 200 J → large fish 20 J (2).

OR

(d) What would happen to this pond if all the decomposers were removed? Give two reasons.

> Dead matter and wastes would pile up and the nutrient recycling would stop (1); producers would soon be deprived of nutrients and the ecosystem would collapse (1).

16. Explain how urine is formed in the human kidney. Name the organs through which urine is stored and passed out. How does an artificial kidney help patients whose kidneys have failed?

> Filtration of blood in the glomerulus into Bowman's capsule (1); selective reabsorption of glucose, amino acids, salts and water in the tubule (1); tubular secretion of extra salts/waste and urine collected in the collecting duct (1); ureter → urinary bladder → urethra (1); in haemodialysis the patient's blood passes through tubes in a dialyser bathed in dialysing fluid, nitrogenous wastes diffuse out and the cleaned blood is returned (1).

OR

(i) Draw a labelled diagram of a longitudinal section of a bisexual flower. (ii) Explain the process of fertilisation in flowering plants and state what the zygote and the ovary develop into.

> (i) Labelled diagram showing sepal, petal, stamen (anther, filament), carpel (stigma, style, ovary, ovule) (2). (ii) Pollen lands on the stigma, grows a pollen tube through the style carrying the male gametes to the ovule; one male gamete fuses with the egg to form the zygote (2); the zygote becomes the embryo/seed and the ovary becomes the fruit (1).

SECTION B — CHEMISTRY — 12 questions · 25 marks

Q17–Q21 MCQs. Q22: Assertion (A) / Reason (R) – choose A, B, C or D.

17. A student heats lead nitrate crystals strongly in a dry boiling tube. Reddish-brown fumes are given off and a yellow residue remains. Which of the following correctly describes the change?

A. Thermal decomposition; the fumes are of nitrogen dioxide
B. Displacement reaction; the fumes are of nitrogen
C. Combination reaction; the fumes are of nitrogen dioxide
D. Photolytic decomposition; the fumes are of oxygen

> **Ans: A** $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$ on heating; NO_2 is reddish-brown and the residue PbO is yellow.

18. An iron nail is kept dipped in copper sulphate solution for about 30 minutes. Which of the following observations and inferences is correct?

A. The blue solution turns pale green because iron is more reactive than copper
B. The blue solution turns colourless because copper is more reactive than iron
C. The green solution turns blue because iron is more reactive than copper
D. No change occurs because iron is less reactive than copper

> **Ans: A** $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$: Fe displaces Cu, the blue Cu^{2+} solution becomes pale green Fe^{2+} solution and brown copper deposits.

19. A chemist needs a salt solution that turns phenolphthalein pink. Which of the following salts should she choose?

A. NaCl B. Na_2CO_3 C. NH_4Cl D. KNO_3

> **Ans: B** Na_2CO_3 is the salt of a strong base (NaOH) and weak acid (H_2CO_3) and its solution is basic (pH ≈ 11).

20. An oxide X reacts with dilute hydrochloric acid as well as with sodium hydroxide solution, forming a salt and water in each case. X is most likely:

A. Na_2O B. CuO C. Al_2O_3 D. CO_2

> **Ans: C** Amphoteric oxides react with both acids and bases; Al_2O_3 is amphoteric, Na_2O and CuO are basic, CO_2 is acidic.

21. The molecular formula of the alkene having five carbon atoms (pentene) is:

A. C_5H_{12} B. C_5H_{10} C. C_5H_8 D. C_6H_{12}

Ans: B Alkenes follow C_nH_{2n}, so n = 5 gives C_5H_{10}.

22. **Assertion (A):** Aluminium oxide is called an amphoteric oxide.
Reason (R): It reacts with both acids and bases to give salt and water.

Ans: A R is the definition of an amphoteric oxide and explains A.

23. Write the balanced equation for the reaction between aqueous sodium sulphate and aqueous barium chloride. State the type of the reaction and the colour of the precipitate.

$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4\downarrow + 2NaCl$ (1). Double displacement (precipitation) reaction; white precipitate of barium sulphate (1).

24. Why should an acid always be added to water, and not water to acid, while preparing a dilute solution? What happens to the concentration of H_3O^+ ions on dilution?

Mixing is highly exothermic; adding water to concentrated acid can make the mixture splash out and cause burns (1). On dilution the concentration of H_3O^+ ions per unit volume decreases (1).

25. Show the formation of magnesium chloride ($MgCl_2$) from magnesium and chlorine using electron-dot structures. Name the type of bond and state its nature.

Mg (2, 8, 2) loses two electrons to form Mg^{2+} (2, 8); each Cl (2, 8, 7) gains one electron to form Cl^- (2, 8, 8) (1½ with correct dot structure). Ionic (electrovalent) bond formed by transfer of electrons (1); electrostatic attraction between ions (½).

OR

Give reasons: (a) Sodium is stored under kerosene. (b) Aluminium utensils resist corrosion although aluminium is a reactive metal. (c) Copper does not react with dilute hydrochloric acid.

(a) Sodium reacts violently with oxygen and moisture of air (1). (b) A thin protective layer of Al_2O_3 forms on the surface (1). (c) Copper is below hydrogen in the reactivity series, so it cannot displace hydrogen from the acid (1).

26. Distinguish between saturated and unsaturated hydrocarbons giving one example of each. Name one chemical test to distinguish between them.

Saturated: only single bonds, e.g. ethane C_2H_6 (1). Unsaturated: contain at least one double or triple bond, e.g. ethene C_2H_4 (1). Unsaturated hydrocarbons decolourise bromine water (brown) / alkaline $KMnO_4$ whereas saturated ones do not (1).

27. In a laboratory, Sneha warms a mixture of ethanol and ethanoic acid with a few drops of concentrated sulphuric acid in a water bath for a few minutes and then pours the mixture into a beaker of water. A sweet, fruity smell is noticed. Later she adds a pinch of sodium carbonate to a test tube of ethanoic acid and observes brisk effervescence.

(a) Name the compound responsible for the sweet smell and the class of compounds it belongs to.

Ethyl ethanoate; an ester (½ + ½).

(b) Write the chemical equation for its formation. What is the role of concentrated H_2SO_4?

$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$ (½); conc. H_2SO_4 acts as a dehydrating agent/catalyst (½).

(c) Name the gas evolved with sodium carbonate and write the balanced equation of the reaction.

Carbon dioxide (½); $2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$ (1½).

OR

(d) Name the reaction that converts an ester back to an alcohol and a sodium salt of the acid by NaOH. State one use of this reaction.

> Saponification (1); used in the manufacture of soap (1).

28. Define the following types of chemical reactions and give a balanced equation for each: (a) thermal decomposition, (b) photolytic decomposition, (c) electrolytic decomposition, (d) displacement, (e) double displacement.

> (a) Breakdown by heat: $CaCO_3 \rightarrow CaO + CO_2$ (1). (b) Breakdown by light: $2AgCl \rightarrow 2Ag + Cl_2$ (1). (c) Breakdown by electricity: $2H_2O \rightarrow 2H_2 + O_2$ (1). (d) A more reactive element displaces a less reactive one: $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (1). (e) Exchange of ions: $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$ (1). (½ for definition, ½ for equation; any alternate valid equation accepted.)

OR

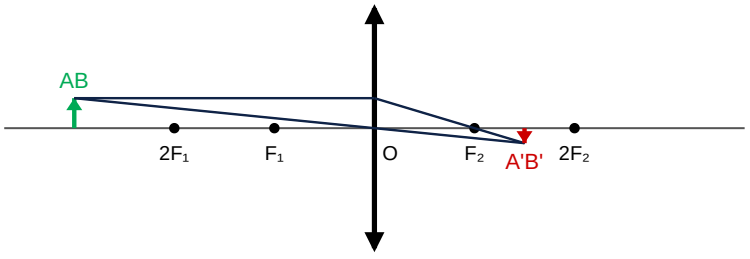
Describe the chlor-alkali process: name the electrolyte, the products formed at each electrode and at the end, give the balanced equation and write one use of each product.

> Electrolysis of aqueous NaCl (brine) (1). Cathode: H_2; anode: Cl_2; NaOH remains in solution (1). $2NaCl + 2H_2O \rightarrow 2NaOH + Cl_2 + H_2$ (1). Uses: H_2 – fuel/margarine; Cl_2 – water treatment/bleach; NaOH – soaps, paper (any one for each, 2).

SECTION C — PHYSICS — 11 questions · 25 marks

Q29–Q32 MCQs. Q33: Assertion (A) / Reason (R) – choose A, B, C or D.

29. A candle flame is placed 30 cm from a convex lens of focal length 10 cm (see figure). The image formed is:



A. real, inverted and diminished, between F_2 and $2F_2$

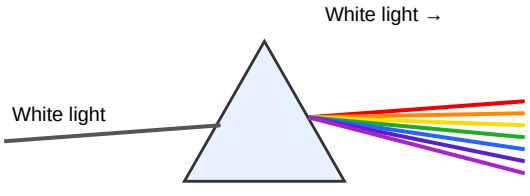
B. real, inverted and magnified, beyond $2F_2$

C. virtual, erect and magnified

D. real, inverted and of the same size at $2F_2$

> **Ans: A** u = 30 cm = 3f; v = uf/(u − f) = 15 cm, between F_2 (10 cm) and $2F_2$ (20 cm); m = −0.5.

30. White light is passed through a glass prism as shown in the figure. The colour of the spectrum that is deviated the most is:

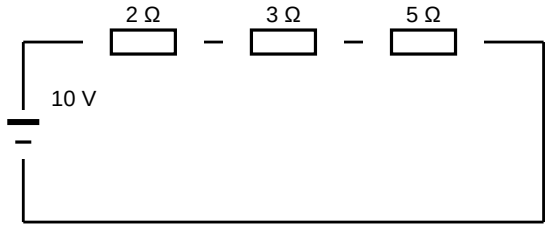


A. red B. yellow C. green D. violet

> **Ans: D** Violet has the shortest wavelength, the highest refractive index in glass and therefore the greatest

deviation.

31. Three resistors of 2 Ω, 3 Ω and 5 Ω are connected in series with a 10 V battery as shown. The potential difference across the 5 Ω resistor is:



A. 1 V B. 2.5 V C. 5 V D. 10 V

Ans: C R = 10 Ω, I = 1 A, V = IR = 1 × 5 = 5 V.

32. In India, the AC supply has a frequency of 50 Hz. The direction of the current reverses after every:

A. 1/50 s B. 1/100 s C. 1/25 s D. 1/200 s

Ans: B One cycle takes 1/50 s and contains two reversals, so the current reverses every 1/100 s.

33. **Assertion (A):** A fuse wire is made of an alloy having a low melting point.
Reason (R): The heating effect of excess current melts the fuse wire and breaks the circuit.

Ans: A R explains why a low melting point is required.

34. Two thin lenses of power +3.5 D and −2.5 D are placed in contact with each other. Find the power and the focal length of the combination.

$P = P_1 + P_2 = 3.5 - 2.5 = +1$ D (1). $f = 1/P = 1$ m = 100 cm (convex combination) (1).

OR

The refractive index of glass with respect to air is 1.5. Define the term refractive index and find the speed of light in glass (speed of light in air = 3×10^8 m/s).

Refractive index is the ratio of the speed of light in vacuum/air to its speed in the medium (1). $v = c/n = 2 \times 10^8$ m/s (1).

35. A person with hypermetropia can see clearly objects at 50 cm but cannot read at the normal near point of 25 cm. Find the power and nature of the lens required for her to read at 25 cm.

The lens must form a virtual image at 50 cm of an object at 25 cm: u = −25 cm, v = −50 cm (1). 1/f = 1/v − 1/u = −1/50 + 1/25 = 1/50 → f = +50 cm (1). P = +2 D; convex lens (1).

36. An electric bulb is rated 55 W, 220 V. Calculate (i) the current drawn, (ii) its resistance and (iii) the energy consumed in kWh when it is used for 4 hours.

(i) I = P/V = 55/220 = 0.25 A (1). (ii) R = V/I = 220/0.25 = 880 Ω (1). (iii) E = 0.055 kW × 4 h = 0.22 kWh (1).

37. A straight wire carries current from the south to the north. (a) State the rule used to determine the direction of the magnetic field around it. (b) How does the strength of the field change as we move away from the wire? (c) What happens to the direction of the field if the current is reversed?

(a) Right-hand thumb rule: holding the wire with the right hand, thumb along the current, the curled fingers give the direction of the field (1). (b) The field decreases with the distance from the wire (1). (c) The direction of the field also reverses (1).

38. Priya uses a convex lens of power +5 D as a magnifying glass to look at a small insect on a leaf. She holds the lens such that the insect is 10 cm from the lens and sees a large upright image of the insect.

(a) Calculate the focal length of the lens.

> f = 1/P = 1/5 m = 20 cm (1).

(b) Why does she see a large upright image? Name the nature of the image.

> The object is between the optical centre and the focus (u < f); the image is virtual, erect and magnified (1).

(c) Find the position of the image and the magnification produced.

> u = −10, f = +20: 1/v = 1/20 − 1/10 = −1/20 → v = −20 cm (1); m = v/u = +2 (1).

O R

(d) At what distance from the lens should the insect be placed to obtain a real image of the same size? Justify your answer.

> u = 2f = 40 cm (1); the image is then formed at 2f on the other side, real, inverted and of the same size (m = −1) (1).

39. An object 2 cm high is placed 15 cm in front of a concave mirror of focal length 10 cm. Find the position, nature and size of the image. Draw the ray diagram. State one use of a concave mirror.

> u = −15 cm, f = −10 cm: 1/v = 1/f − 1/u = −1/10 + 1/15 = −1/30 → v = −30 cm (1½); m = −v/u = −2 (1); h′ = −4 cm (½). Image real, inverted, magnified, 30 cm in front of the mirror (1). Ray diagram (1). Use e.g. shaving mirror/headlight/solar furnace (not required if diagram given ⇒ ½ + ½).

O R

(a) What is meant by electric short-circuiting and overloading? (b) Name the colour of the live, neutral and earth wires. (c) State the functions of the earth wire and of the fuse in a domestic circuit.

> (a) Short-circuit: live and neutral wires touch, resistance becomes nearly zero and a very large current flows; overloading: too many appliances on one circuit draw more current than its rating (2). (b) Live – red, neutral – black, earth – green (1½). (c) Earth wire gives a low-resistance path to ground, preventing shocks from metal bodies of appliances; the fuse melts and breaks the circuit when current exceeds the safe value (1½).